

DAR ES SALAAM INTL, Tanzania, United Republic of

WMO: 638940

Lat: 6.878S

Lon: 39.203E

Elev: 182

StdP: 14.60

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	65.0	66.2	60.2	78.3	80.2	62.2	84.4	78.4	22.8	82.1	21.1	82.7	1.6	200	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	92.3	79.4	91.4	79.0	90.2	78.3	81.7	89.3	80.8	87.9	80.0	86.7	12.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	153.1	85.6	79.0	151.3	85.1	78.5	148.9	84.4	45.6	89.9	44.6	88.6	43.8	87.7	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	18.8	17.6	61.2	95.0	3.0	1.1	59.0	95.8	57.3	96.5	55.6	97.1	53.3	97.9
			59.2	83.7	3.2	1.3	56.9	84.7	55.1	85.5	53.3	86.2	51.0	87.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.0	83.5	83.5	82.9	80.6	79.0	77.2	75.9	76.3	77.4	79.3	81.0	83.0	(6)
	DBStd	3.33	1.97	1.97	2.12	2.08	1.73	1.73	1.67	1.76	1.67	1.96	1.96	2.02	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	10932	1040	939	1019	917	899	815	804	816	821	909	930	1023	(10)
	CDD65	5457	575	519	554	467	434	365	339	351	371	444	480	558	(11)
	CDH74	50368	6494	5882	5819	3887	3164	2490	2281	2483	2978	3939	4729	6222	(12)
	CDH80	17755	2410	2315	2268	1175	941	732	621	732	1018	1463	1728	2352	(13)
(14) Wind	WSAvg	7.6	11.3	9.7	6.0	5.7	6.8	7.3	7.9	6.9	6.8	6.9	6.6	9.0	(14)
(15) Precipitation	PrecAvg	44.1	2.6	2.3	5.6	10.4	6.6	1.4	0.9	0.9	1.0	2.6	4.7	4.5	(15)
	PrecMax	68.2	10.6	8.7	16.8	22.4	16.0	6.1	6.1	4.2	3.8	11.2	16.7	14.8	(16)
	PrecMin	28.5	0.0	0.0	1.0	0.5	0.6	0.0	0.0	0.0	0.1	0.0	0.0	0.0	(17)
	PrecStd	10.9	2.3	2.0	3.1	4.4	3.9	1.5	1.1	0.9	0.9	2.4	4.3	3.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.9	93.5	93.6	91.4	89.6	88.1	87.7	88.2	89.7	91.4	91.7	92.8	(19)
		MCWB	79.6	79.7	80.7	79.5	76.8	74.9	73.5	73.7	74.2	76.2	77.6	79.9	(20)
	2%	DB	91.3	91.8	91.8	89.7	87.9	86.8	86.2	86.8	88.0	89.7	90.0	91.2	(21)
		MCWB	79.0	79.2	80.0	79.1	76.6	74.4	73.0	73.2	73.7	75.9	77.4	78.9	(22)
	5%	DB	89.9	91.0	91.1	88.2	86.6	85.9	85.0	85.8	86.8	88.2	89.3	89.8	(23)
		MCWB	78.3	78.9	79.7	78.7	76.4	73.9	72.6	72.7	73.3	75.5	77.3	78.5	(24)
	10%	DB	89.1	89.6	89.6	86.4	85.6	84.3	83.9	84.3	85.8	87.3	87.9	88.9	(25)
		MCWB	78.2	78.2	79.1	78.4	76.2	73.4	72.2	72.4	73.0	75.3	77.2	78.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.7	82.2	83.6	81.9	79.6	77.9	76.8	76.6	77.8	79.3	80.1	81.7	(27)
		MCDB	89.9	91.5	91.4	88.2	85.7	85.5	84.7	85.4	86.1	87.0	87.4	89.7	(28)
	2%	WB	80.4	80.8	81.9	80.7	78.7	76.6	75.3	75.3	76.2	78.1	79.3	80.5	(29)
		MCDB	88.1	88.9	89.3	87.0	84.7	83.7	83.4	83.5	84.5	86.0	86.2	87.6	(30)
	5%	WB	79.6	79.8	80.8	79.8	77.9	75.7	74.3	74.4	75.1	77.1	78.8	79.8	(31)
		MCDB	86.5	87.0	87.5	85.6	83.7	82.5	82.1	82.2	83.3	85.0	85.4	86.4	(32)
	10%	WB	79.0	79.1	79.9	79.0	77.1	74.8	73.4	73.5	74.2	76.3	78.1	79.2	(33)
		MCDB	85.3	85.8	86.1	84.2	82.8	81.3	80.7	81.0	82.1	83.7	84.5	85.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	13.4	14.2	12.4	13.4	15.9	17.2	17.7	18.4	16.8	14.6	12.7	(35)
		MCDBR	13.1	15.0	15.8	15.0	15.4	17.4	18.6	19.2	19.7	18.3	16.0	13.8	(36)
	5% WB	MCWBR	4.5	5.2	5.7	6.1	6.2	6.9	7.5	7.5	7.1	6.5	5.4	4.7	(37)
		MCWBR	12.1	13.6	14.6	13.5	13.6	15.7	17.5	17.8	18.0	16.3	14.0	12.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.480	0.485	0.432	0.401	0.402	0.397	0.404	0.406	0.405	0.407	0.407	0.442	(40)
		taud	2.248	2.243	2.429	2.529	2.504	2.487	2.432	2.417	2.413	2.425	2.457	2.383	(41)
	Ebn at Noon		274	273	283	284	275	271	271	279	287	291	293	284	(42)
		Edh at Noon	47	47	38	33	33	32	35	37	38	39	37	41	(43)
(44) All-Sky Solar Radiation	RadAvg		1854	1899	1761	1516	1456	1489	1527	1631	1807	1906	1916	1880	(44)
	RadStd		124	110	121	134	128	60	54	49	65	70	90	106	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page